# Simpler Model Link Seed and Growth Directly

## Data

We now have a time-series across individuals $i = 1 \ldots, I$ in year $t = 1, \ldots, T$:

- $sc_{i,t} \in \mathbb{N}$ : seed count of tree $i$ in year $t$
- $\mathrm{BAI}_{i,t} \geq 0$ : basal area increment of tree $i$ in year $t$
- $\mathrm{DBH}_{i,t} \geq 0$ : DBH of tree $i$ in year $t$
- $\mathrm{GST}_t \in \mathbb{R}$ : growing season temperature in year $t$

## Parameters

### Growth parameters

- $\alpha_{\mathrm{BAI}} \in \mathbb{R}$ : intercept for growth
- $\beta_{\mathrm{GST1}} \in \mathbb{R}$ : effect of climate on growth
- $\sigma_{\mathrm{BAI}} > 0$ : standard deviation for growth

### Reproduction parameters

- $\alpha_{\mathrm{sc}} \in \mathbb{R}$ : intercept for seeds
- $\beta_{\mathrm{dbh}} \in \mathbb{R}$ : effect of tree size on seeds
- $\beta_{\mathrm{GSt2}} \in \mathbb{R}$ : effect of climate on seeds
- $\gamma_{\mathrm{current}} \in \mathbb{R}$ : current year's growth effect on seeds
- $\gamma_{\mathrm{lag}} \in \mathbb{R}$ : lag effect, last year's growth effect on seeds
- $\phi_{\mathrm{sc}} > 0$ : dispersion parameter

## Prior Distributions

$$\alpha_{\mathrm{BAI}} \sim \mathcal{N}(0, 5)$$

$$\beta_{\mathrm{GST1}} \sim \mathcal{N}(0, 5)$$

$$\sigma_{\mathrm{BAI}} \sim \mathcal{N}(0, 1)$$

$$\alpha_{\mathrm{sc}} \sim \mathcal{N}(0, 5)$$

$$\beta_{\mathrm{dbh}} \sim \mathcal{N}(0, 5)$$

$$\beta_{\mathrm{GST2}} \sim \mathcal{N}(0, 5)$$

$$\phi_{sc} \sim \mathrm{Gamma}(2, 0.1)$$

$$\gamma_{\mathrm{current}} \sim \mathcal{N}(0, 5)$$

$$\gamma_{\mathrm{lag}} \sim \mathcal{N}(0, 5)$$

## Model

For each individual tree $i$ and year $t$:

Expected log-growth for all $t = 1, \ldots, T$:

$$G_{i,t} = \alpha_{\mathrm{BAI}} + \beta_{\mathrm{GST1}} \cdot \mathrm{GST}_t$$

Observed (BAI):

$$\mathrm{BAI}_{i,t} \sim \mathrm{LogNormal}\left(G_{i,t},\, \sigma_{\mathrm{BAI}}\right)$$

Expected seed count for all $t = 1, \ldots, T$:

$$\log(\mu_{\mathrm{sc},i,t}) = \alpha_{\mathrm{sc}} + \beta_{\mathrm{dbh2}} \cdot \mathrm{DBH}_{i,t} + \beta_{\mathrm{GST2}} \cdot \mathrm{GST}_t + \gamma_{\mathrm{current}} \cdot G_{i,t} + \gamma_{\mathrm{lag}} \cdot G_{i,t-1}$$

Observed seed counts:

$$sc_{i,t} \sim \mathrm{NegBinomial2Log}\left(\log(\mu_{\mathrm{sc},i,t}),\, \phi_{\mathrm{sc}}\right)$$